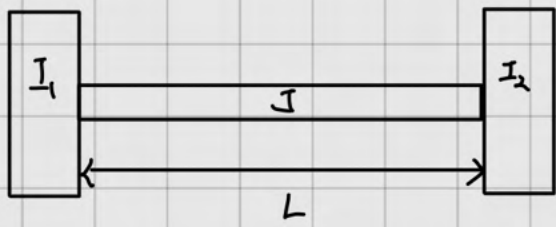


1. 질량관성모멘트가 각각 I_1, I_2 인 두 개의 원판이 길이 L , 면적극관성모멘트 J 인 균일한 축 양끝에 고정되어 있다. 이 비틀림 진동 시스템에 대한 특성방정식을 구하라.



비틀림에 관한 식이네

$$\frac{\partial^2 \theta}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 \theta}{\partial t^2} \quad c = \sqrt{\frac{G}{J}}$$

이때 좌단과 우단에 질량관성모멘트 I_1, I_2 가 있어

$\Sigma M = J \alpha^2$, 양 끝단에서의 작용하는 토크가 질량관성 모멘트의 각가속도를 발생시켜

$$GJ \frac{\partial \theta}{\partial x} \Big|_{x=0} = I_1 \frac{\partial^2 \theta}{\partial t^2} \Big|_{x=0}$$

$$GJ \frac{\partial \theta}{\partial x} \Big|_{x=L} = I_2 \frac{\partial^2 \theta}{\partial t^2} \Big|_{x=L}$$

$$\left. \begin{aligned} GJ X'(0) T(t) &= I_1 X(0) T''(t) \\ GJ X'(L) T(t) &= I_2 X(L) T''(t) \end{aligned} \right\} \rightarrow \frac{X'(0)}{X'(L)} = \frac{I_1 X(0)}{I_2 X(L)}$$

특성방정식

I, L

$$\theta(x, 0) = \theta_0(x)$$

$$\dot{\theta}(x, 0) = \dot{\theta}_0(x)$$

$$\theta(x, t) = X(x) T(t) \text{ 라고 하면}$$

$$\frac{\partial^2 \theta}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 \theta}{\partial t^2}$$

$$X''(x) T(t) = \frac{1}{c^2} X(x) T''(t)$$

$$\frac{X''(x)}{X(x)} = \frac{1}{c^2} \frac{T''(t)}{T(t)} = -\sigma^2 \text{ 라고 하면}$$

$$X'' + \sigma^2 X = 0$$

$$X = a_1 \cos \sigma x + a_2 \sin \sigma x$$

$$X' = -\sigma a_1 \sin \sigma x + \sigma a_2 \cos \sigma x$$

경계조건을 적용하면

$$\frac{a_2}{-a_1 \sin \theta + a_2 \cos \theta} = \frac{I_1}{I_2} \times \frac{a_1}{a_1 \cos \theta + a_2 \sin \theta}$$

$$a_2 (a_1 \cos \theta + a_2 \sin \theta) = a_1 \times \frac{I_1}{I_2} (-a_1 \sin \theta + a_2 \cos \theta)$$

$$\left(a_2^2 + a_1^2 \times \frac{I_1}{I_2} \right) \sin \theta = \left(a_1 a_2 \times \frac{I_1}{I_2} - a_1 a_2 \right) \cos \theta$$

$$\tan \theta = \frac{\left(a_1 a_2 \times \frac{I_1}{I_2} - a_1 a_2 \right)}{\left(a_2^2 + a_1^2 \times \frac{I_1}{I_2} \right)}$$

$$\theta = \frac{1}{I} \tan^{-1} \left(\frac{\left(a_1 a_2 \times \frac{I_1}{I_2} - a_1 a_2 \right)}{\left(a_2^2 + a_1^2 \times \frac{I_1}{I_2} \right)} \right)$$

θ를 얻는다.

2. 균일한 막대의 끝이 $Y \sin \omega t$ 의 운동을 받고 있다.
이 막대 종진동의 정상상태 응답 $w(x, t)$ 을 구하라.

균일한 단면 보에서

크기, 속, λ_0, v_0

$$\frac{\partial^2 W}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 W}{\partial t^2} \quad \left(c = \sqrt{\frac{E}{\rho}} \right)$$

B.C.

$$W(0, t) = 0$$

자유단에서 힘이 0

$$EA \frac{\partial W}{\partial x} \bigg|_{x=L} = 0$$

$$W(x, t) = X(x) T(t) \text{ 라고 가정}$$

$$\frac{X''(x)}{X(x)} = \frac{T''(t)}{c^2 T(t)} = -\beta^2$$

$$X'' + \beta^2 X = 0$$

$$X(x) = a_1 \cos \beta x + a_2 \sin \beta x$$

B.C. 조건을 생각하면

$$a_1 = 0$$

$$EA X'(L) T(t) = 0$$

$$X'(L) = 0$$

$$\beta a_2 \cos \beta L = 0$$

$$\therefore \beta_n = \frac{(2n-1)\pi}{2L}$$

$$\therefore X_n(x) = a_n \sin \frac{(2n-1)\pi}{2L} x$$

$$\omega_n = \beta_n c$$

$$c^2 X''(x) T(t) - X(x) \ddot{T}(t) = -Y \sin \omega t$$

$$X(x) (\ddot{T}(t) + \omega_n^2 T(t)) = Y \sin \omega t$$

$$\int_0^L \sin^2 \frac{(2n-1)\pi}{2L} x \, dx (\ddot{T}(t) + \omega_n^2 T(t)) = Y \sin \omega t \int_0^L \sin \frac{(2n-1)\pi}{2L} x \, dx$$

$$\frac{1}{2} (\ddot{T}(t) + \omega_n^2 T(t)) = 0$$

$$\therefore \ddot{T}(t) + \omega_n^2 T(t) = 0$$

$$T(t) = B_n \cos(\omega_n t + \phi_n)$$

$$B_n = \frac{\sqrt{\omega_n^2 x_0^2 + v_0^2}}{\omega_n}$$

$$\phi_n = \tan^{-1} \left(\frac{\omega_n x_0}{v_0} \right)$$

$$\therefore W(x, t) = X_n(x) T_n(t)$$

$$= \sum_{n=1}^{\infty} \sin \beta_n x$$

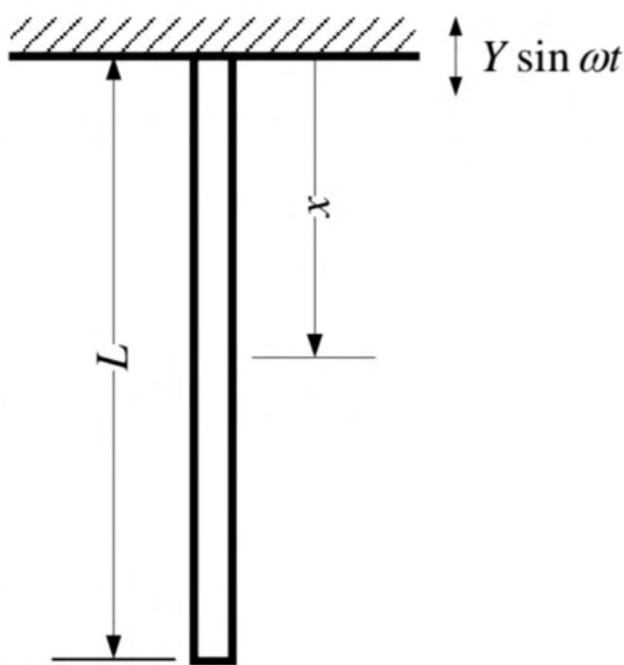
기타 계산은 13 섹션

$$\beta_n = \frac{(2n-1)\pi}{2L} \quad c = \sqrt{\frac{E}{\rho}}$$

$$\omega_n = \beta_n c = \frac{(2n-1)\pi}{2L} \sqrt{\frac{E}{\rho}}$$



$$\left(\frac{\sqrt{\omega_n^2 x_0^2 + v_0^2}}{\omega_n} \cos(\omega_n t + \tan^{-1} \left(\frac{\omega_n x_0}{v_0} \right)) \right)$$



3. 양 끝이 핀 고정된(단순지지) 균일한 보의 굽힘진동에 대한 고유진동수와 모드형상을 구하라. 처음 세 모드의 형상을 플롯하라.

$$\frac{\partial^4 w}{\partial x^4} + \frac{1}{c^2} \frac{\partial^2 w}{\partial t^2} = 0 \quad c = \sqrt{\frac{EI}{\rho A}}$$

B.C.	I.C.
$w(0,t) = 0$	$w(x,0) = 0$
$w(l,t) = 0$	$w_t(x,0) = \dot{w}_0(0)$
$EI \frac{\partial^2 w}{\partial x^2} \Big _{x=0} = 0$	
$EI \frac{\partial^2 w}{\partial x^2} \Big _{x=l} = 0$	

$$w(x,t) = X(x)T(t) e^{i\omega t}$$

$$c^2 \frac{X''(x)}{X(x)} = -\frac{T''(t)}{T(t)} = \omega^2$$

$$c^2 \frac{X''(x)}{X(x)} = -\omega^2$$

$$X''''(x) - \frac{\omega^2}{c^2} X(x) = 0$$

$$X''''(x) - \beta^4 X(x) = 0 \quad \beta^4 = \frac{\omega^2}{c^2}$$

$$X = Ae^{\beta x} + B e^{-\beta x} + C \sin \beta x + D \cos \beta x$$

$$X(x) = a_1 \sin \beta x + a_2 \cos \beta x + a_3 \sinh \beta x + a_4 \cosh \beta x$$

B.C. 적용

$$X(0) = 0 = a_2 + a_4$$

$$X(l) = 0 = (a_1 \sin \beta l + a_2 \cos \beta l + a_3 \sinh \beta l + a_4 \cosh \beta l) \times \rho^2$$

$$EI \frac{\partial^2 w}{\partial x^2} \Big|_{x=0} = 0 = EI \times (-a_2 + a_4) \rightarrow \therefore a_2 = a_4 = 0$$

$$EI \frac{\partial^2 w}{\partial x^2} \Big|_{x=l} = 0 = (-a_1 \sin \beta l + a_3 \sinh \beta l) \times \rho^2 \times EI$$

$$EI \beta^2 a_1 \sin \beta l = EI \beta^2 a_3 \sinh \beta l$$

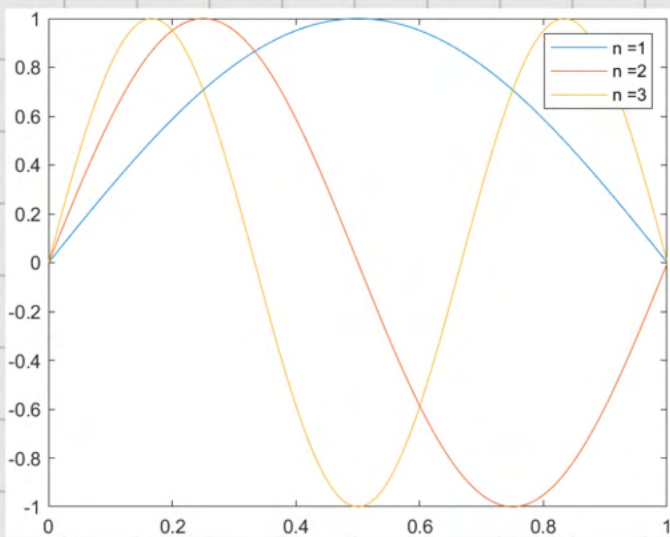
$$\beta^2 a_1 \sin \beta l = -\beta^2 a_3 \sinh \beta l$$

$$a_3 = 0 \quad \therefore \sin \beta l = 0$$

$$\beta l = n\pi \quad (n=0,1,\dots)$$

$$\beta = \frac{n\pi}{l} \quad \rightarrow \quad \therefore \omega = \beta^2 \sqrt{\frac{EI}{\rho A}} = \frac{(n\pi)^2}{l^2} \sqrt{\frac{EI}{\rho A}}$$

$$\therefore X(x) = \sin \frac{n\pi}{l} x \quad \text{모드 형상}$$



4. 양 끝이 자유로운 균일한 보의 굽힘진동에 대한 고유진동수와 모드형상을 구하라. 처음 세 모드의 형상을 플롯하라. (강제모드는 무시하라.)

$$\frac{\partial^4 w}{\partial x^4} + \frac{1}{c^2} \frac{\partial^2 w}{\partial t^2} = 0 \quad c = \sqrt{\frac{EI}{\rho A}}$$

B.C.

$$EI \frac{\partial^2 w}{\partial x^2} \Big|_{x=0} = 0$$

$$EI \frac{\partial^2 w}{\partial x^2} \Big|_{x=l} = 0$$

$$\frac{\partial}{\partial x} \left(EI \frac{\partial^2 w}{\partial x^2} \right) \Big|_{x=0} = 0$$

$$\frac{\partial}{\partial x} \left(EI \frac{\partial^2 w}{\partial x^2} \right) \Big|_{x=l} = 0$$

I.C.

$$w(x,0) = 0$$

$$w_t(x,0) = \dot{w}_0(x)$$

$$w(x,t) = X(x)T(t) \text{ 사용}$$

$$c^2 \frac{X''''(x)}{X(x)} = - \frac{T''(t)}{T(t)} = \omega^2$$

$$c^2 \frac{X''''(x)}{X(x)} = \omega^2$$

$$X''''(x) - \frac{\omega^2}{c^2} X(x) = 0$$

$$X''''(x) - \beta^4 X(x) = 0 \quad \left(\beta^4 = \frac{\omega^2}{c^2} \right)$$

$$X = A e^{\beta x} + B e^{-\beta x} + C \sin \beta x + D \cos \beta x$$

$$X(x) = a_1 \sin \beta x + a_2 \cos \beta x + a_3 \sinh \beta x + a_4 \cosh \beta x$$

$$X'(x) = \beta^2 \left(-a_1 \sin \beta x - a_2 \cos \beta x + a_3 \sinh \beta x + a_4 \cosh \beta x \right)$$

$$X'''(x) = \beta^3 \left(-a_1 \cos \beta x + a_2 \sin \beta x + a_3 \cosh \beta x + a_4 \sinh \beta x \right)$$

B.C. 적용하면

$$X'(0) = 0 = \beta^2 (-a_2 + a_4) \rightarrow a_2 = a_4$$

$$X'''(0) = 0 = \beta^3 (-a_1 + a_3) \rightarrow a_1 = a_3$$

$$X'(l) = 0 = \beta^2 (-a_1 \sin \beta l - a_2 \cos \beta l + a_3 \sinh \beta l + a_4 \cosh \beta l)$$

$$X'''(l) = 0 = \beta^3 (-a_1 \cos \beta l + a_2 \sin \beta l + a_3 \cosh \beta l + a_4 \sinh \beta l)$$

$$\begin{bmatrix} \sinh \beta l - \cosh \beta l & \cosh \beta l - \cos \beta l \\ \cosh \beta l - \cos \beta l & \sinh \beta l + \sin \beta l \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = 0$$

$$\det A = 0 \text{ 이어야 함}$$

$$\sinh^2 \beta l - \sin^2 \beta l - \cosh^2 \beta l + 2 \cosh \beta l \cos \beta l - \cos^2 \beta l = 0$$

$$2 \cosh \beta l \cos \beta l = 2$$

$$\cosh \beta l \cos \beta l = 1$$

$$\cos \beta l = \frac{1}{\cosh \beta l}$$

고유진동수

$$\therefore \omega_n = \frac{(2n+1)^2 \pi^2}{4l^2} \sqrt{\frac{EI}{\rho A}} \quad (n \geq 5)$$

$$\omega_n = \left(\frac{4.73}{l} \right)^2 \sqrt{\frac{EI}{\rho A}} \quad n < 5 \text{ 일 때는 언어린 값을}$$

$$\vdots \quad \quad \quad \text{극한 대입.}$$

$$\therefore \beta_n l \approx \frac{2n+1}{2} \pi \quad (n=5, 6, \dots)$$

$$\therefore \beta_n \approx \frac{(2n+1)\pi}{2l}$$



$$\beta_n l = 4.73$$

$$7.85$$

$$10.99$$

$$14.13$$

$$17.21$$

$$\frac{(2n+1)\pi}{2} \quad (n \geq 5 \text{ 인 경우})$$

이고 B.C를 대입한 X 이 $\beta_n l$ 값을 대입해 정리하면

모드형상은 $X(x)$ 는 다음과 같이 주어진다.

$$\cosh \beta_n x + \cos \beta_n x - G_n (\sinh \beta_n x + \sin \beta_n x)$$

$$G_n = \frac{\cosh \beta_n l - \cos \beta_n l}{\sinh \beta_n l - \sin \beta_n l}$$

$l=1$ 이며
모드형상을 그리면

